reaches your eyes alongside the white ray $\mathcal{A}$. The white ray brings you an image of a bit of the thin ring; the black ray, an image of a bit of the shadow's edge. The shell of fire is responsible for merging the rays side by side and directing them toward your eyes.

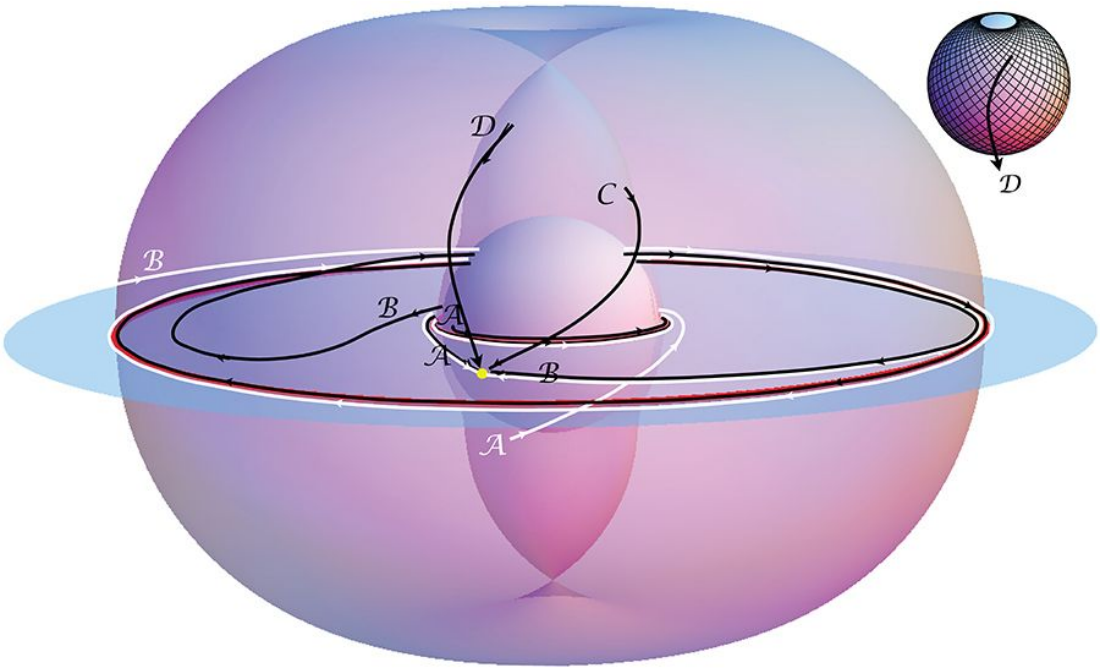


Fig. 8.2. Gargantua (*central spheroid*), its equatorial plane (blue), its shell of fire (*purple and violet*), and black and white light rays that bring you images of the shadow's edge and the thin ring alongside it.

Similarly for the white and black rays $\mathcal{B}$, except they get trapped on the outer edge of the shell of fire going clockwise (struggling against the whirl of space), while rays $\mathcal{A}$ are trapped on the inner edge going counterclockwise (and driven by the whirl of space). The flattening of the shadow's left edge in Figure 8.1 and rounding of its right edge are due to rays $\mathcal{A}$ (left edge) coming from the inner edge of the shell of fire, very close to the horizon, and rays $\mathcal{B}$ (right edge) from the outer edge of the shell of fire, much further out.